universität wien
Faculty of Psychology

Lernen unter COVID-19-Bedingungen

School principals

Method Report

1. Introduction

The present cross-sectional study is the second phase of the project "Learning under COVID-19 conditions" with pupils and students previously archived at https://doi.org/10.11587/VRQL3B and at https://doi.org/10.11587/XIU3TX, respectively.

The second phase of the project comprises three additional target groups – parents, teachers and school principals, in one cross-sectional measurement point. Thus, this phase of the project aims to investigate behaviour and coping of other agents in the school system, beside pupils', in order to get better picture in the situation and effects of COVID-19 on education system in Austria.

Study Description

1.1. Title and version

Lernen unter COVID-19-Bedingungen: SchulleiterInnen (Learning under COVID-19 conditions: school principals, V1.0)

1.2. Principal Investigators

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1.3. Funding/Acknowledgments

The present study has been funded by the Vienna Science and Technology Fund (WWTF), by MEGA Bildungsstiftung, and the City of Vienna through project COV20-025.

1.4. Fieldwork agency

The fieldwork of the study "Lernen unter COVID-19-Bedingungen" was managed and handled internally.

1.5. Topics/Keywords

- Sociodemographic information: age, sex, gender role identity, school type, region, number of teachers and students in school
- Leadership in crisis: self-efficacy, overload, leadership styles, subjective performance, challenges, support
- Contact with authorities: communication patterns and information

2. Study Design

2.1. Sampling

The sample of the study "Lernen unter COVID-19-Bedingungen" consisted of school principals/headmasters working in secondary education in Austria. The principals were invited to participate in the survey by project team members. Moreover, the link to and a recommendation to participate in the survey was published by the Federal Ministry of Education, Science, and Research on their website. The link with a respective invitation to the study was also placed on the survey's website. In order to participate in the study, the respondents had to agree to the informed consent, be Austrian residents, and be older than 18.

2.2. Fieldwork

The overall data collection was conducted in the time from 06 April 2021 to 15 May 2021.

3. Data Cleaning

3.1. General data cleaning procedure

The cleaning procedure contained two steps:

3.1.1. Initial data cleaning

As participating school principals/headmasters answered an online questionnaire via Unipark (Questback, 2007), data was exported from unipark in .csv format and imported to SPSS. Variables names were replaced. All missing values were defined as -99. Invalid zero values or values of -77 were replaced with -99. Participants who gave consent and had answering time (variable duration) larger than zero were selected for a new SPSS file.

3.1.2. Data cleaning

As unipark assigns -1 to variable duration in cases where the break was made during the answering the questionnaire, all valid -1 cases were replaced with (+) 9999999 value. Participants with 100% missings on main outcome variables (related to self-efficacy) were excluded. Minimum possible duration was tested by filling in the questionnaire as fast as possible while reading all items. All cases above the duration treshold and above 18 years og age were selected for a new SPSS file.

3.1.3. Missing values

All empty spaces were replaced with -99. Missing values that were represented by 0 in the raw data set are replaced by -99 on all variables where 0 is not a valid value. Missing values represented with -77 in unipark were replaced with -99 as well. Missing values on variables that were result of filters applied, were defined with -98. Cases where

participants by mistake got to see a question they should not have seen (due to technical issues in unipark) were replaced with -78.

3.1.4. Filters

Data records of participants who did not give their consent are deleted.

A fastest possible time for answering the questionnaire is determined diligently. Variable **duration** of answers with duration below the fastest possible time is set to 9999999.

Data records with 100% missings on main outcome variables are deleted.

All other filters are defined in the questionnaire, as well as in data set using values -98 and -78 (see above) as value labels.

3.1.5. Answers to open questions

All open questions have been removed from the data sets due to anonymization rules.

3.1.6. Multiple choice / single choice

Questions "Welche Schulart/en leiten Sie?" (Volksschule Mittelschule (MS), Gymnasium (AHS), Berufsbildende höhere Schule (BHS), Berufsbildende mittlere Schule (BMS), Berufsschule und Lehre, Polytechnische Schule (PTS), Fachmittelschule (FMS)), "In welchen Bundesländern liegen die Schulen, die Sie leiten?" (Burgenland, Kärnten, Niederösterreich, Oberösterreich, Salzburg, Steiermark, Tirol, Vorarlberg, Wien), "Wird an Ihrer Schule/Ihren Schulen spezielle Deutschförderung angeboten?" (nein; ja, Deutschförderkurse; ja, Deutschförderklassen; ja, integrierte Deutschförderung; ja, andere) were posed in multiple choice format.

3.2. Anonymisation procedure

3.2.1. Code variables

Due to the high risk for identification, variables **es.code** (personal code) and **es.code.school** (school code) were deleted from the data set.

3.2.2. Federal state variables

As there were fewer than 20 observations for "ausgewählt" category of **es.school.fs1** (Burgenland) and **es.school.fs9** (Wien), it was merged with **es.school.fs3** (Niederösterreich) to make (Niederösterreich/Burgenland/Wien). As there were fewer than 20 observations for "ausgewählt" category of **es.school.fs2** (Kärnten), it was merged with **es.school.fs6** (Steiermark) to make (Steiermark/Kärnten). As there were fewer than 20 observations for "ausgewählt" category of **es.school.fs7** (Tirol) and **es.school.fs8** (Vorarlberg), it was merged with **es.school.fs5** (Salzburg) to make (Salzburg/Tirol/Vorarlberg).

3.2.3. Age variables

Data containing the age of the participants in this dataset (es.age) are categorized by aggregating age cohorts to get groups with at least 20 observations.

3.2.4. School type variables

As there were fewer than 20 observations for "ausgewählt" category of the school type variables es.school.type2 (Mittelschule (MS)), es.school.type6 (Berufsschule und Lehre), es.school.type7 (Polytechnische Schule (PTS)), es.school.type8 (Fachmittelschule (FMS)), es.school.type9 (Sonderschule), and es.school.type10 (Andere) they were removed from the dataset.

3.2.5. Number of schools, teachers and students

As there were fewer than 20 observations for "2 oder mehr" category of the number of schools managed by principals (es.nr.schools), the variable was removed from the dataset.

Data containing the number of teachers in schools managed by principals (es.teachers.nr) are categorized by aggregating the data to get groups with at least 20 observations.

Data containing the number of students in schools managed by principals (es.pupils.nr) are categorized by aggregating the data to get groups with at least 20 observations.

3.2.6. Variables related to German courses

As there were fewer than 20 observations for "ausgewählt" category of es.ger.class3 (Deutschförderung: ja, Deutschförderklassen) and es.ger.class5 (Deutschförderung: ja, andere), they were removed from the dataset.

3.2.7. Open questions

In this data set, 9 open variables are deleted without recoding or standardizing the answers:

- es.school.type.other (Welche Schulart/en leiten Sie? Andere)
- es.ger.class.other (Wird an Ihrer Schule/Ihren Schulen spezielle Deutschförderung angeboten? ja, andere:)
- es.ger.class.change (Wie haben sich die Möglichkeiten zur schulischen Deutschförderung seit Beginn der Pandemie verändert?)
- es.accomplishment2 (Sie hatten in den letzten Monaten die Aufgabe, Ihre Schule möglichst gut durch die Covid-Krise zu führen. Wie zufrieden sind Sie damit, wie gut Ihnen das gelungen ist? Platz für Erläuterungen:)
- es.challeneges (Was waren für Sie als Schulleiter/in die größten Herausforderungen seit Beginn der Pandemie?)
- es.accomplish (Was ist Ihnen als Schulleiter/in seit Beginn der Pandemie besonders gut gelungen?)
- es.ils (Was waren aus Ihrer Sicht die wichtigsten Elemente Ihres Führungsstils während der Covid-Krise?)

- es.ils.change2 (Welche Veränderungen waren nötig?)
- es.support (Gibt es Unterstützungsangebote, die für Sie als Schulleitung derzeit besonders hilfreich wären? Wenn ja, welche?)

3.2.8. Open questions containing numeric values

Answers to questions requiring numeric values are recoded according to the description in section "Data cleaning procedure". Original (string) answers are deleted.

3.3. Categorizing variables

Variables containing numeric values in connection with time and numbers of people were categorized into suitable categories that allow analyses among different points of measurement.

3.4. Further deletions

3.4.1. Consent variable

One variable containing the participants' agreement on processing the filled in data for scientific purposes is deleted in each measurement point (es.consent) as all included participants gave their consent for processing the filled in data for scientific purposes.

3.4.2. Gender variables

For es.gender, as no one selected "divers" or "möchte ich nicht sagen" categories, these two categories were removed from the available categories.

3.4.3. Filter variables

One filter variable (filter_$) originating from the cleaning procedure is deleted in each measurement point.

3.4.4. Variables with few valid values

In this dataset, one variable that contains only missing values (quality) is deleted.

3.4.5. System variables

System variables that contain information of the duration of answering separate pages are deleted.

- external_lfdn (external sequence number)
- tester (filter variable indicating entries made for test purposes)
- dispcode (disposition code)
- lastpage (identifier of exit page)
- browser (browser identifier)

- referer (HTTP referer)
- page_history (array of page identifiers indicating the sequence of visited questionnaire pages)
- hflip (questionnaire answered on a screen in landscape mode)
- vflip (questionnaire answered on a screen in portrait mode)
- session_id (session identifier)
- Language (language)
- ats (absolute timestamp)
- datetime (date of first access)
- date_of_last_access (date of last access)
- date_of_first_mail (date of first e-mail)
- rts4985945, rts4985949, rts4985951, rts4985952, rts5009020, rts5009025, rts5009106, rts5030539, rts5030552, rts5030629 (relative timestamps for pages)

3.4.6. Reverse-coded variables

All reverse-coded variables are deleted.

- es.ils1.rec (Reversed Likert scale of item es.ils1)
- es.ils2.rec (Reversed Likert scale of item es.ils2)
- es.ils3.rec (Reversed Likert scale of item es.ils3)
- es.ils4.rec (Reversed Likert scale of item es.ils4)
- es.ils5.rec (Reversed Likert scale of item es.ils5)
- es.ils6.rec (Reversed Likert scale of item es.ils6)
- es.ils7.rec (Reversed Likert scale of item es.ils7)
- es.ils8.rec (Reversed Likert scale of item es.ils8)
- es.ils9.rec (Reversed Likert scale of item es.ils9)
- es.ils10.rec (Reversed Likert scale of item es.ils10)
- es.ils11.rec (Reversed Likert scale of item es.ils11)
- es.ils12.rec (Reversed Likert scale of item es.ils12)
- es.ils13.rec (Reversed Likert scale of item es.ils13)
- es.ils14.rec (Reversed Likert scale of item es.ils14)
- es.tls1.rec (Reversed Likert scale of item es.tls1)
- es.tls2.rec (Reversed Likert scale of item es.tls2)
- es.tls3.rec (Reversed Likert scale of item es.tls3)
- es.tls4.rec (Reversed Likert scale of item es.tls4)
- es.tls5.rec (Reversed Likert scale of item es.tls5)
- es.tls6.rec (Reversed Likert scale of item es.tls6)
- es.tls7.rec (Reversed Likert scale of item es.tls7)
- es.info1.rec (Reversed Likert scale of item es.info1)

- es.info2.rec (Reversed Likert scale of item es.info2)
- es.info3.rec (Reversed Likert scale of item es.info3)
- es.info4.rec (Reversed Likert scale of item es.info4)
- es.accomplishment1.rec (Reversed Likert scale of item es.accomplishment1)
- es.m1.rec (Reversed Likert scale of item es.m1)
- es.m2.rec (Reversed Likert scale of item es.m2)
- es.m3.rec (Reversed Likert scale of item es.m3)
- es.m4.rec (Reversed Likert scale of item es.m4)
- es.m5.rec (Reversed Likert scale of item es.m5)
- es.m6.rec (Reversed Likert scale of item es.m6)
- es.f1.rec (Reversed Likert scale of item es.f1)
- es.f2.rec (Reversed Likert scale of item es.f2)
- es.f3.rec (Reversed Likert scale of item es.f3)
- es.f4.rec (Reversed Likert scale of item es.f4)
- es.f5.rec (Reversed Likert scale of item es.f5)
- es.f6.rec (Reversed Likert scale of item es.f6)
- es.workh.rec (Reversed Likert scale of item es.workh)
- es.wol1.rec (Reversed Likert scale of item es.wol1)
- es.wol2.rec (Reversed Likert scale of item es.wol2)
- es.wol3.rec (Reversed Likert scale of item es.wol3)
- es.wol4.rec (Reversed Likert scale of item es.wol4)
- es.wol5.rec (Reversed Likert scale of item es.wol5)

3.4.7. Mean scale variables

All variables that contain calculated mean values for each scale are deleted.

- es.ils.mean (Mean scale Instructional leadership (reversed items used))
- es.tls (Mean scale Transformational leadership (reversed items used))
- es.ag (Mean scale Agency (reversed items used))
- es.co (Mean scale Communion (reversed items used))
- es.info (Mean scale Information from Schulbehörden (reversed items used))
- es.wol (Mean scale Work overload (reversed items used))
- es.eff (Mean scale Efficacy for management)
- es.effils (Mean scale Efficacy for instructional leadership)

3.5. Increasing reusability

To increase the reusability of the data set variables are rearranged into an appropriate order. Variable names are renamed to get more meaningful names.